

AI-Based Error Extraction & Personalized Learning Path Analysis

1. AI Rubric-Based Error Extraction (Advanced Component 1)

1.1. How the AI Extracts Errors According to the IELTS Rubric:

The AI will evaluate IELTS Writing Task 1 and Task 2 essays based on the four official IELTS band descriptors:

Criterion	What It Measures
Task Achievement (TA)	How well the essay addresses the prompt and fulfills the task requirements
Coherence & Cohesion (CC)	Logical organization, paragraph structure, and use of linking words
Lexical Resource (LR)	Vocabulary range, precision, spelling, and word choice
Grammatical Range & Accuracy (GRA)	Sentence variety, grammar correctness, and punctuation

Extraction Process:

1. The learner submits an essay through the platform.
2. The system sends the essay to a Large Language Model (LLM) with a structured prompt that includes:
 - The full IELTS band descriptors for each criterion.
 - Instructions to analyze the essay line-by-line.
 - A requirement to output errors mapped to specific rubric categories.
3. The AI returns a structured response containing:
 - Identified errors grouped by rubric criterion.
 - Suggested corrections for each error.
 - A provisional band score for each of the four criteria.
 - Source references (for instance, "Based on IELTS public band descriptor for Task Achievement").

Example Output:

Error	Rubric Criterion	Correction
"The number of student increased"	Grammatical Range	"The number of students increased"
Overuse of "increase" (4 times)	Lexical Resource	Use synonyms: "rise," "grow," "climb"
No clear overview in introduction	Task Achievement	Add a sentence summarizing the main trend

1.2. How Accuracy Will Be Validated

To ensure the AI's grading is reliable, we will:

- Prepare a test set of 20–30 sample IELTS essays with pre-assigned band scores from official sources (e.g., Cambridge IELTS books, British Council samples).
- Run these essays through the AI and record its band score predictions.
- Compare AI scores against the human-assigned scores.
- Calculate the Mean Absolute Error (MAE) across all four criteria. A MAE of ≤ 0.5 band points will be considered acceptable.
- Manually review cases where the AI's score differs significantly from the human score to identify patterns of error.

This validation will be documented and included in the interim and final reports.

2. Personalized Learning Path Algorithm (Advanced Component 2)

2.1. How the Algorithm Suggests a Personalized Path

The system recommends learning content based on each learner's historical performance and weakness profile.

Input Data:

- Scores from completed quizzes (Reading, Listening).
- AI-graded essay scores across the four rubric criteria.
- Lesson completion status and time spent on each module.

Recommendation Logic (Rule-Based Approach):

The system will identify the learner's lowest-scoring rubric criterion and recommend targeted content accordingly:

Weakest Criterion	Recommended Action
Task Achievement	Suggest lessons on "Understanding Prompts" and "Essay Structure"
Coherence & Cohesion	Suggest lessons on "Paragraphing" and "Linking Words"
Lexical Resource	Suggest vocabulary-building exercises and "Synonyms for Task 1"
Grammatical Range	Suggest grammar modules

Example:

- A learner scores TA: 6.5, CC: 6.0, LR: 5.5, GRA: 6.0.
- The system detects LR (5.5) as the weakest.
- It recommends: "Vocabulary Building for Describing Trends" and "Avoiding Repetition in Task 1."

2.2. How the Personalized Path Will Be Evaluated

We will compare the personalized approach against a static learning path (a fixed sequence of lessons that all learners follow).

Evaluation Plan:

- Use a simulated learner dataset with varied performance profiles (e.g., weak in grammar, weak in vocabulary, well-balanced).
- Simulate two scenarios for each learner:
 - Scenario A: Follows the static path.
 - Scenario B: Follows the personalized path.
- Measure:
 - Improvement rate (increase in band scores over time).
 - Time to completion (how quickly the learner finishes the recommended modules).

- Document the results and show that personalized paths lead to faster or more targeted improvement.

3. Sample Data Sources for AI Validation

To validate the AI's grading accuracy, we will use the following data sources:

Source	Description
Cambridge IELTS Official Practice Books	Published sample essays with official band scores. Reliable gold-standard data.
British Council / IDP Sample Answers	Free online resources with annotated band descriptors.
Public IELTS Essay Datasets	Limited, but may provide additional test cases (e.g., Kaggle, UCI repositories).
Human-Reviewed AI-generated Essays	We will generate additional essays using LLMs and have a human expert (English teacher) assign provisional band scores.

Data Usage Summary:

- 20–30 essays will be reserved for final accuracy testing.
- Additional essays will be used for prompt engineering and tuning the AI's grading instructions.
- All data will be used strictly for academic evaluation and will not be shared outside the project.